Latest Advances in Prestressed Concrete Construction

Abstract:

Prestressed concrete (PSC) is very important in modern construction. Today, new changes are happening in this field because of the need for sustainable and smart construction methods. This paper explains three main developments: use of advanced materials, smart monitoring systems, and automated construction techniques.

In the past, steel tendons in PSC structures were damaged by corrosion, which reduced the life of structures. Now, new materials like Carbon Fiber Reinforced Polymer (CFRP) and Basalt Fiber Reinforced Polymer (BFRP) are being used. These materials do not rust and are stronger and lighter than steel.

Modern PSC structures also use Internet of Things (IoT) sensors to monitor performance. These sensors give real-time information about stress and structural health, helping engineers maintain structures before problems occur. In addition, technologies like 3D concrete printing and robotic tendon placement reduce human errors and save labour.

Overall, these advancements make prestressed concrete structures stronger, more durable, environmentally friendly, and easier to manage.